Contents

Chapter 1

Introduction to Data Visualization

It is difficult for the human mind to look at a list of numbers and identify the patterns in them, so we often use these numbers to make a picture. These pictures are called *graphs*, *charts*, or *plots*. Often, the right picture can make the meaning in the data obvious. **Data visualization** is the process of making pictures from numbers.

1.1 Types of Variables

Depending on the type of data and what you are trying to demonstrate about it, you will use different types of data visualizations. Before we get to the more complex types of data visualizations, let's look at the two types of variables we are usually looking to visualize: *categorical* and *numerical*.

Categorical or qualitative variables are those that can be put into categories. For example, the type of shoes a person wears (sneakers, sandals, etc.) is a categorical variable. The type of shoes sold is a categorical variable. The name of the salesperson (John, Jane, etc.) is also a categorical variable.

Numerical or quantitative variables are those that can be measured with numbers. For example, the number of boxes of cookies sold is a quantitative variable. The average temperature on a given day is also a quantitative variable. Quantitative variables can be further divided into *discrete* and *continuous* variables. Discrete variables are those that can only take on specific values, such as the number of pencils sold (you can't sell 3.5 pencils). Continuous variables can take on any value within a range, such as the average height of students in a class (which can be 5.5 feet, 1.65 m, etc.).

Further, we can also classify data into univariate, bivariate, and multivariate data. Univariate data has only one variable, such as the number of boxes of cookies sold. Bivariate data has two variables, such as the average temperature and total sales for a lemonade stand. Multivariate data has more than two variables, such as the average temperature, total sales, and type of ookie sold. We will only look at univariate and bivariate data!

1.2 How do we describe data?

There are different descriptive methods for representing data, and they can usually be grouped into three categories: *tabular*, *graphical*, and *numerical*.

Tabular methods include tables, which are a way of organizing data into rows and columns. They are useful for displaying raw data and making it easy to look up specific values. However, they can be difficult to interpret when dealing with large datasets or when trying to identify patterns.

Graphical methods include bar charts, line graphs, pie charts, and scatter plots. They are useful for visualizing data and making it easier to identify patterns and trends.

Numerical methods include measures of central tendency (mean, median, mode) and measures of variability (range, variance, standard deviation). They are useful for summarizing data and providing a numerical representation of the data's characteristics.

We will focus on graphical methods in this chapter, but we will also touch on tabular and numerical methods in later chapters.

1.3 Univariate Data

We will start with univariate data, which has only one variable. For example, the number of boxes of cookies sold by each salesperson is univariate data. Bar charts, line graphs, and pie charts are all good for visualizing univariate, *categorical* data. We will explore these types of graphs in the next sections.

1.3.1 Bar Chart

Bar charts are typically used to show the values of a categorical variable. Each bar represents a category, and the height of the bar represents the value of that category. Bar charts can have vertical or horizontal bars to represent this data.

Here is an example of a bar chart.

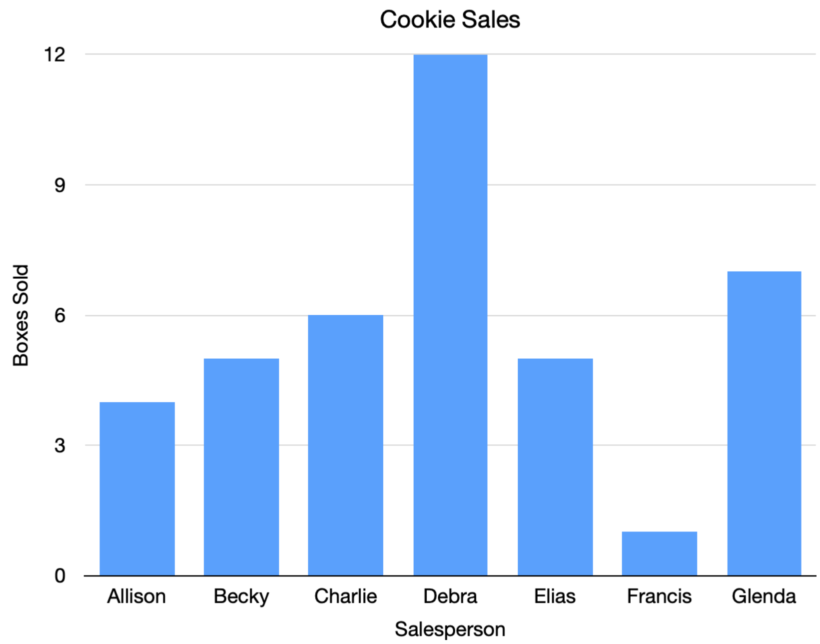


Figure 1.1: A bar chart showing cookie sales per person.

Each bar represents the cookie sales of one person. For example, Charlie has sold 6 boxes of cookies, so the bar goes over Charlie's name and reaches to the number 6.

Looking at this chart, you probably think, "Wow, Debra has sold a lot more cookies than anyone else, and Francis has sold a lot fewer."

The same data could be in a table like this:

Salesperson	Boxes Sold
Allison	4
Becky	5
Charlie	6
Debra	12
Elias	5
Francis	1
Glenda	7

A table (especially a large table) is often just a bunch of numbers. A chart helps our brains understand what the numbers mean.

Bar charts can also go horizontally.

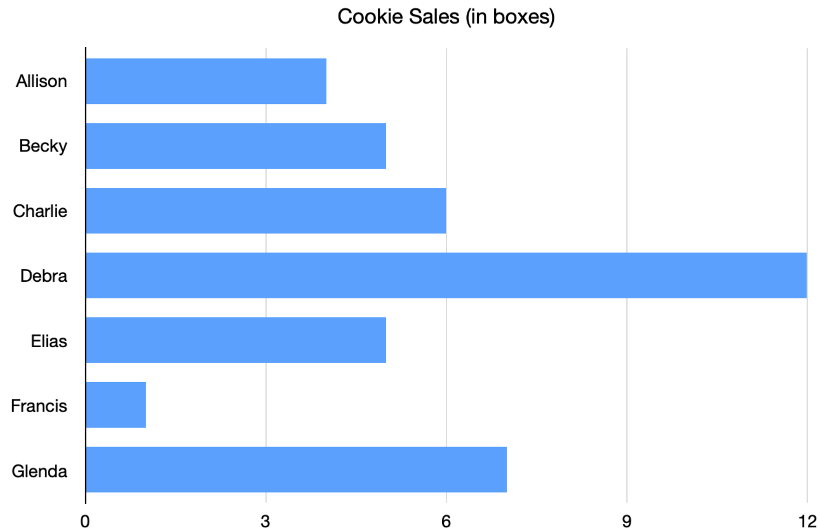


Figure 1.2: A horizontal bar chart showing the same cookie sales data.

Sometimes we use colors to explain what contributed to the number.

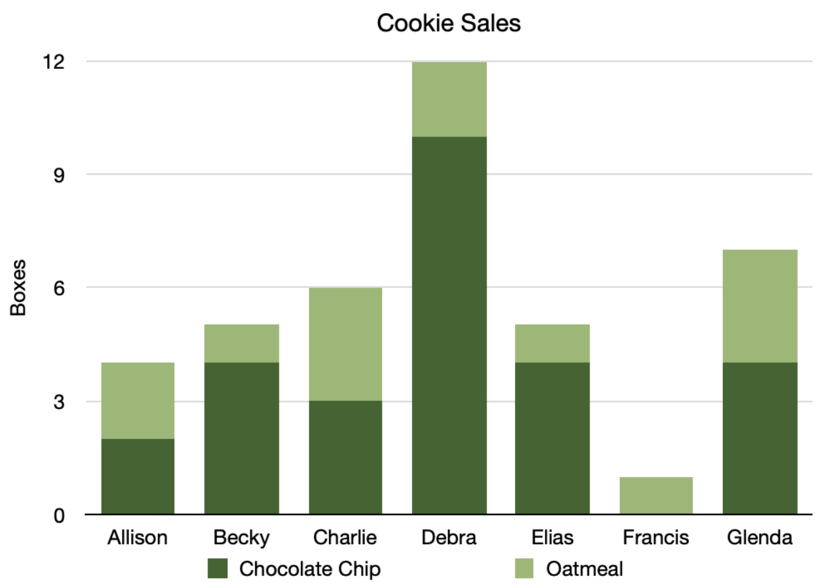


Figure 1.3: A bar chart with different colors showing types of cookies.

This tells us that Becky sold more boxes of chocolate chip cookies than boxes of oatmeal cookies.

Make Bar Graph

Go back to your compound interest spreadsheet and make a bar graph that shows both balances over time:

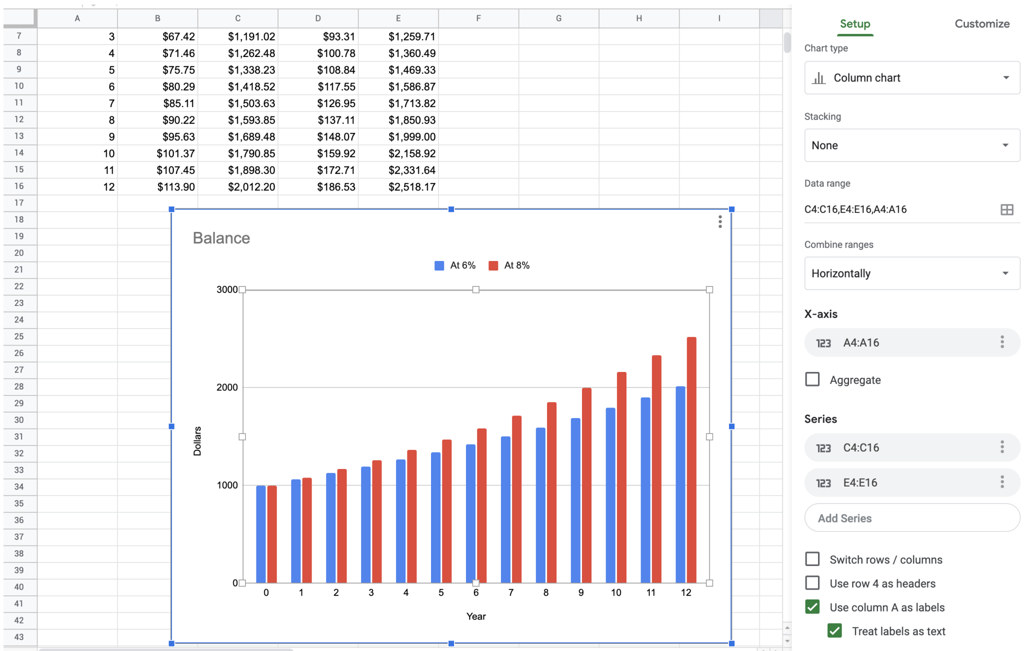


Figure 1.4: A bar graph made in google sheets showing interest.

The year column should be used as the x-axis. There are two series of data that come from C4:C16 and E4:E16. Tidy up the titles and legend as much as you like. Looking at the graph, you can see the balances start the same, but balance of the account with the larger interest rate quickly pulls away from the account with the smaller interest rate.

1.3.2 Line Graph

Here is a line graph:

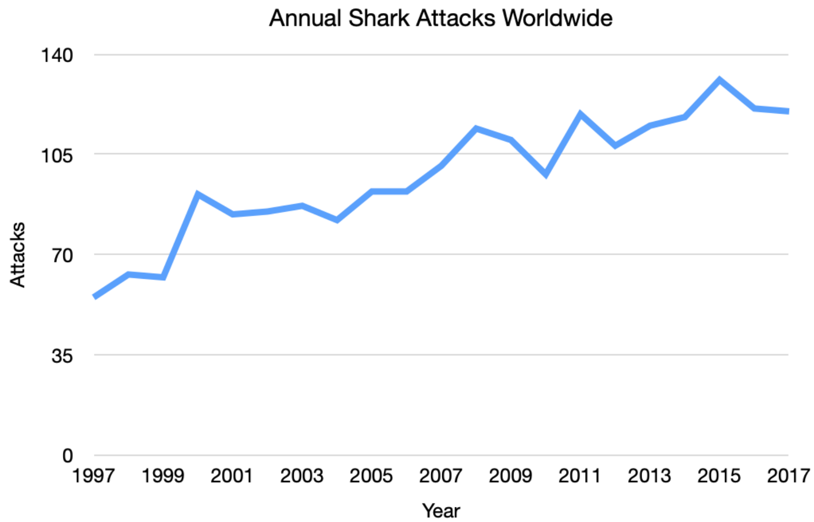


Figure 1.5: A line graph showing shark attacks per year over two decades.

These are often used to show trends over time. Here, for example, you can see that the number of shark attacks has been increasing over time.

You can have more than one line on a graph.

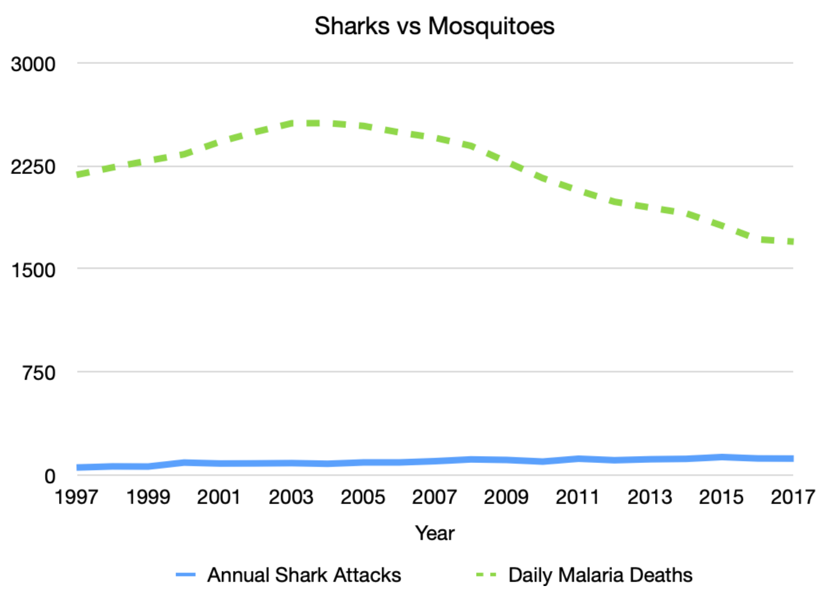


Figure 1.6: A line graph showing shark deaths versus mosquito deaths.

1.3.3 Pie Chart

You use a pie chart when you are looking at the comparative size of numbers. This is best for comparing percentages of a whole that sum to 100%. Here we can see that Nitrogen makes up 78% of the gases in the air.

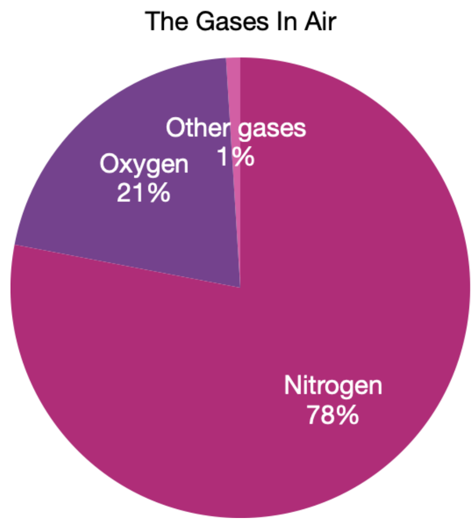


Figure 1.7: A pie chart of the various gases in the air.

When to use a Pie Chart Make a Pie Chart in Python

For *quantitative*, univariate data, we often use dotplots and stemplots to visualize smaller sets of data; and histograms, cumulative frequency plots, and boxplots to visualize larger sets of data. We will explore these types of graphs below.

1.3.4 Dotplot

A *dotplot* is one of the simplest ways to display single, quantitaive variables. Each observation is represented by a dot placed above its value on a number line. If a value appears more than once, the dots stack vertically.

Dotplots are most effective for small to moderate data sets. If the data set is too large, a dotplot can become cluttered and hard to read. In that case, a histogram or a boxplot may be a better choice.

When to Use a Dotplot

- You have 1 variable.

- The variable is quantitative.
- You want to see the distribution and the individual data values.
- The data set is not too large.

How to Make a Dotplot

1. Draw a horizontal number line (x-axis) that covers the data range.
2. Choose a scale that fits the entire range of values.
3. Place one dot above the location of each observation.
4. If multiple observations have the same value, stack the dots vertically.

Here is an example of a dotplot.

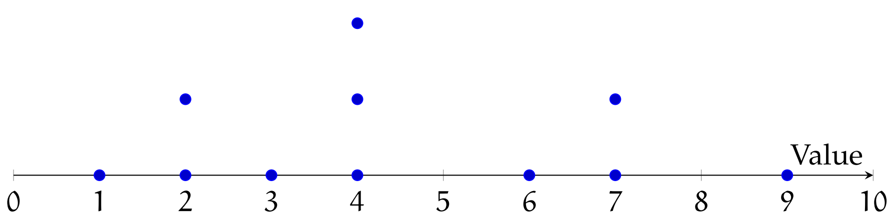


Figure 1.8: A dotplot showing a small sample distribution.

How to Read a Dotplot

Each dot shows the location of one data value. To find the value of a dot, look straight down to the number line.

A dotplot helps you describe the distribution by answering questions like:

- Spread: How spread out are the values (range and clustering)?
- Shape: Do the dots form a roughly symmetric shape, or is the distribution skewed?
- Center: Where is the distribution "balanced" (a typical value)?
- Outliers: Are there values far away from the rest?

Sometimes a dotplot shows a distribution with a long tail to one side. For example, salary data often has many values in the lower-to-middle range and a few extremely high values. This creates a right-skewed distribution.

Outliers can strongly affect numerical summaries, especially the mean.

Connecting Outliers and the Mean (Seesaw Idea) A useful way to think about the mean is as a balancing point. Imagine the dotplot as a seesaw: points far from the main cluster have more "pull." So a single extreme value can shift the mean noticeably, even if most data values stay the same.

Comparative Dotplots

A *comparative dotplot* uses the same numerical scale to compare two (or more) groups. This is useful when you want to compare distributions side-by-side.

When comparing groups, focus on differences in:

- center (typical value),
- spread,
- shape (skew/clusters),
- and outliers.

Make a Dotplot in a Spreadsheet

You can create a dotplot in Excel or Google Sheets using a scatter plot and a helper column.

1. Put your data values in column A (starting in A2). Sort them (recommended).
2. In cell B2, enter:
   `=COUNTIF($A$2:A2, A2)`
   Fill down. This creates stacking heights for repeated x-values.
3. Insert a Scatter chart using column A as x-values and column B as y-values.
4. Format the chart:
   - markers only (no lines),
   - hide the y-axis labels (y is only for stacking),
   - add titles and adjust the x-axis scale.

1.3.5 Stemplot

A *stemplot* (also called a *stem-and-leaf plot*) is an effective way to summarize **one quantitative variable** when the data set is not too large. Like a dotplot, it helps you see the overall distribution, but a stemplot has an extra advantage: it still shows the exact data values.

To make a stemplot, each observation is split into two parts:

- the stem, which is the left-most part of the number (leading digit(s))
- the leaf, which is the remaining part, usually the final digit

For example, 23 can be written as 2 | 3.

How to Make a Stemplot

1. Choose stems and leaves. Decide where to split each number into a stem and a leaf. There is no single rule for this. Your goal is to choose a reasonable number of stems.
2. Draw a vertical divider. Stems go on the left; leaves go on the right.
3. List all stems in order. Make sure the stems cover the full range of your data.
4. Write leaves for each observation. Place each leaf next to its correct stem.
5. Sort leaves. Put the leaves for each stem in increasing order.
6. Include a key. The key shows how to interpret the stems and leaves.

Choosing the Number of Stems

If you use *too few stems*, you can hide patterns because the data get lumped together. If you use *too many stems*, you can also hide patterns because the plot becomes too spread out.

A common strategy is to start with stems that represent groups of 10 (tens digits as stems), and then adjust if needed.

Sometimes a single stem represents too wide of an interval. In that case, you can split stems by repeating each stem twice:

- the first row holds leaves 0–4
- the second row holds leaves 5–9

For example, if your data run from 10 to 49, you can use stems

$$1, 1, 2, 2, 3, 3, 4, 4$$

where the first 1 represents 10–14 and the second 1 represents 15–19, and so on.

Example

Suppose these are housing costs (in dollars per month):

$$255, 259, 304, 306, 309, 317, 323, 327, 332, 334, 342, 349, 350, 351, 362, 367, 376, 384, 401, 403$$

If we use the hundreds digit as the stem and the last two digits as the leaf, we can split each stem into two rows: (top row for leaves 00–49, bottom row for leaves 50–99)

Stem	Leaves
2	55 59
2	
3	04 06 09 17 23 27 32 34 42 49
3	50 51 62 67 76 84
4	01 03
4	

Key: 3 | 04 = 304 dollars per month

How to Read a Stemplot

A stemplot gives you several kinds of information at once:

- Exact values: Each leaf represents a single observation.
- Frequencies: The number of leaves in a row tells how many observations fall in that interval. For example, if a stem row has 9 leaves, that means 9 observations are in that range.
- Gaps: If a stem has no leaves, that means there are no observations in that interval.
- Shape and spread: You can see whether the values cluster in certain ranges and whether the distribution is skewed.
- Center: You can estimate a typical value by locating where the leaves cluster most.
- Outliers: Unusually high or low values may stand out because they appear far from the rest.

Just like with dotplots, you can imagine the distribution as something that could "balance" on a finger: the place it would balance is a rough sense of the center.

Make a Stem Plot in a Spreadsheet Make a Stem Plot in Python

1.3.6 Histogram

A *histogram* is one of the most popular ways to display single quantitaive variables. It is especially useful for showing patterns in larger data sets, where dotplots or stemplots can become overwhelming. A histogram is like a stemplot turned on its side: instead of listing individual values, we group values into intervals called classes or bins, and draw a bar for each bin.

Histograms are best when you care about the overall shape of a distribution (clusters, skew, gaps, outliers), not the exact value of every data point.

A histogram can be drawn using frequency (counts), relative frequency (proportions), and percent frequency (percentages). These three versions have the same shape if they use the same bins. The difference is only the y-axis scale.

How to Make a Histogram

1. Create groups from continuous data. This process is called binning.
2. Draw the x-axis and y-axis, and scale them to fit the class intervals and the frequencies (or relative frequencies or percentages).
3. Draw one bar for each class. The height of each bar equals the frequency (or relative frequency or percent) for that class.
4. Draw the bars touching with no gaps (because the data values are continuous and the bins connect).

How to Read a Histogram

- Each bar represents a single class (bin). There is one bar per class.
- The bins are placed on the x-axis in increasing order, like a number line.
- The height of a bar tells how many observations fall in that class (for a frequency histogram).
- A class with no bar means no observations fall in that interval (a gap).

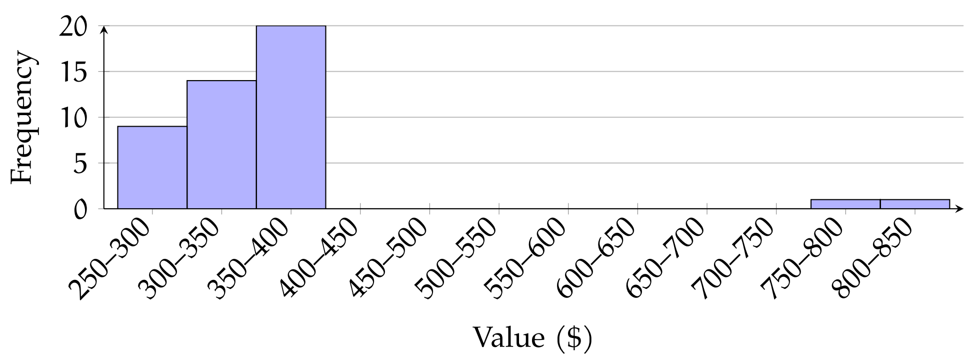


Figure 1.9: A histogram using 50-dollar bins illustrating the example counts (9, 14, 20, ... and 0,1,1 beyond $700).

Example of Interpreting Bars

Suppose a histogram uses the bins \$250–\$300, \$300–\$350, \$350–\$400, and so on.

- If the first bar (250–300) has height 9, then 9 observations are between \$250 and \$300.
- If the second and third bars (300–350 and 350–400) have heights 14 and 20, then $14 + 20 = 34$ observations are between \$300 and \$400.
- If there is no bar for 700–750, then there are 0 observations between \$700 and \$750.
- If the bars beyond 700 have heights 0, 1, and 1, then $0 + 1 + 1 = 2$ observations are at least \$700.

A histogram can help you to describe the distribution by focusing on shape, center, spread, gaps, and outliers.

Making a Histogram in a Spreadsheet

You can make a histogram in Excel or Google Sheets.

1. Put the data in one column.
2. Choose bin widths (for example, bins of width 50: 250–300, 300–350, 350–400, etc.).
3. Insert a histogram chart (or use a column chart after building a frequency table).
4. Check the bin width and the endpoints. Changing the bins can change the appearance of the distribution.

Making a Histogram in Python

Python can generate a histogram quickly and lets you change the bin width easily.

```python
import matplotlib.pyplot as plt

# Replace this list with your data
data = [250, 275, 310, 320, 350, 365, 410, 415, 430, 500, 525, 610, 720, 840]

# Choose the number of bins (try changing this!)
plt.hist(data, bins=8)

plt.title("Histogram")
plt.xlabel("Value")
plt.ylabel("Frequency")
plt.show()
```

Try changing the value of `bins`. Notice that the overall shape can look different depending on how you group the data.

How do we compare dotplots, stemplots, and histograms?

For small data sets, dotplots and stemplots can be better because they show the exact values. In a histogram, you lose that detail because values are grouped into bins. In exchange, histograms make it easier to see the overall pattern in large data sets.

1.3.7 Cumulative Frequency Plot

A *cumulative frequency plot* (also called a *cumulative frequency chart*) shows the running total of how many data values are at or below each value (or at or below each class boundary). Instead of showing how many observations fall in each bin like a histogram, a cumulative frequency plot shows how the totals add up as you move from left to right.

Cumulative frequency plots are useful when you want to answer questions like:

- How many observations are less than or equal to a certain value?
- How many observations are greater than or equal to a certain value?
- What percent of the data fall below a certain cutoff?

How to Make a Cumulative Frequency Plot

1. Draw the x-axis and y-axis.
2. Set up bins for the data and identify the upper class boundaries (the right end-points).
3. Compute the cumulative frequency at each upper class boundary.
   (This means: add the frequency of that class to the total from all previous bins.)
4. Plot a point at each upper class boundary with height equal to the cumulative frequency.
5. Connect the points with straight line segments.

Interpreting the shape of your plot

The steepness of the curve tells you where the data values are concentrated.

- If the curve rises quickly over a certain x-interval, many observations fall in that range.
- If the curve is almost flat, few observations fall in that range.

A cumulative frequency plot can also hint at skewness:

- **Right-skewed:** the curve increases quickly at smaller values, then levels off later.
- **Left-skewed:** the curve increases slowly at first, then rises sharply near the end.
- **Symmetric:** the curve often looks S-shaped.

Example

Suppose $n = 80$ observations were grouped into bins, and the cumulative frequency at $x = 350$ is about 23. Then:

- About 23 observations are ≤ 350.
- About $80 - 23 = 57$ observations are ≥ 350.

The plot below shows a sample cumulative frequency curve. (These coordinates are an example format; you can replace them with your own cumulative totals.)

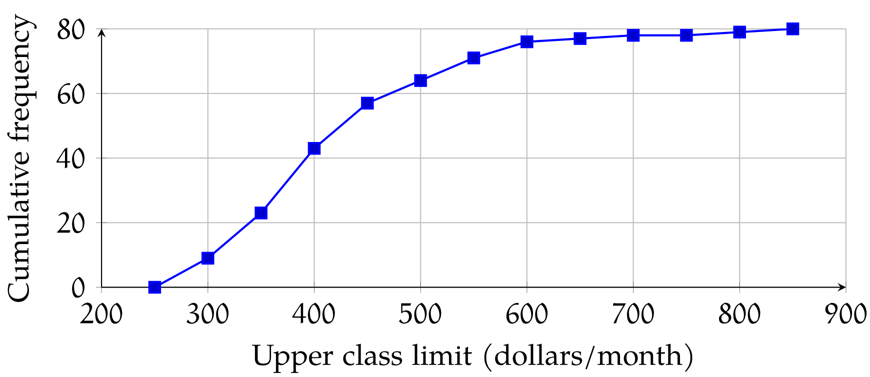


Figure 1.10: A cumulative frequency plot.

Making a Cumulative Frequency Plot in Python

This example shows the basic idea: build bins, count how many values fall in each bin, then take a cumulative sum.

```
import numpy as np
import matplotlib.pyplot as plt

# Replace with your data
data = np.array([250, 275, 310, 320, 350, 365, 410, 415, 430, 500, 525, 610, 720,
    840])

# Choose bin edges (upper class boundaries are the right endpoints)
```

```
bins = np.arange(250, 851, 50)  # 250, 300, ..., 850

# Frequency in each bin
counts, edges = np.histogram(data, bins=bins)

# Cumulative frequency at each upper boundary
cumulative_counts = np.cumsum(counts)
upper_bounds = edges[1:]  # right endpoints

plt.plot(upper_bounds, cumulative_counts, marker='o')
plt.title("Cumulative Frequency Plot")
plt.xlabel("Upper class boundary ($)")
plt.ylabel("Cumulative frequency")
plt.grid(True)
plt.show()
```

If you change the `bin` width, the curve may look different, but it will always increase from left to right and end at n.

Boxplot

1.4 Bivariate Data

1.4.1 Scatter Plot

Sometimes, you have a large number of data points with two values, and you are looking for a relationship between them. For example, maybe you write down the average temperature and the total sales for your lemonade stand on the 15th of every month:

Date	Avg. Temp.	Total Sales
15 January 2022	2.6º C	$183.85
15 February 2022	-4.2º C	$173.56
15 March 2022	13.3º C	$195.22
15 April 2022	26.2º C	$207.61
15 May 2022	27.5º C	$210.88
15 June 2022	31.3º C	$214.18
15 July 2022	33.5º C	$215.23
15 Aug 2022	41.7º C	$224.07
15 September 2022	20.7º C	$198.94
15 October 2022	17.2º C	$196.10
15 November 2022	1.7º C	$185.10
15 December 2022	0.2º C	$188.70

You may wonder, ”Do I sell more lemonade on hotter days?”

To figure this out, you might create a scatter plot. For each day, you put a mark that represents that temperature and the sales that day:

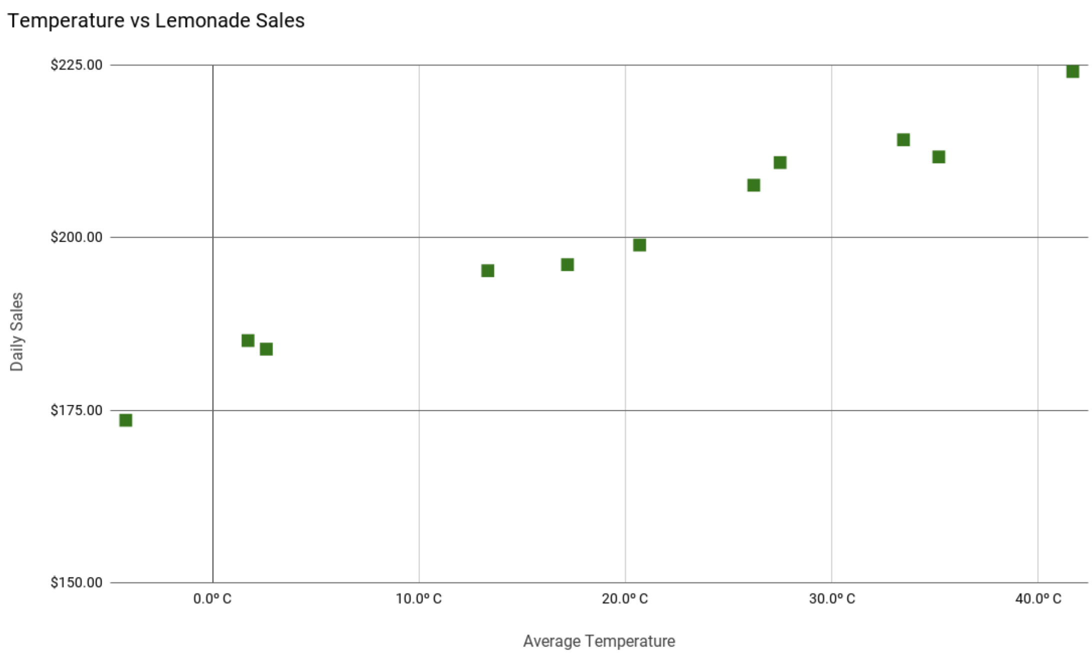


Figure 1.11: A scatterplot of temperature versus daily sales.

From this scatter plot, you can easily see that you do sell more lemonade as the temperature goes up. Drawing a *best-fit* line along the the points will give you a *correlation coefficient*. A positive correlation coefficient will give you a positively proportional relationship, while a negative coefficient will give you a inversely proportional relationship.

Making a Scatter Plot in Python

```python
import matplotlib.pyplot as plt

# Avg. Temp (°C) and Total Sales ($)
temps = [2.6, -4.2, 13.3, 26.2, 27.5, 31.3, 33.5, 41.7, 20.7, 17.2, 1.7, 0.2]
sales = [183.85, 173.56, 195.22, 207.61, 210.88, 214.18, 215.23, 224.07, 198.94,
    196.10, 185.10, 188.70]

plt.scatter(temps, sales)
plt.title("Temperature vs. Lemonade Sales")
plt.xlabel("Average Temperature (°C)")
plt.ylabel("Total Sales ($)")
plt.grid(True)
plt.show()
```

CHAPTER 2

The Dot Product

If you have two vectors $\mathbf{u} = [u_1, u_2, \ldots, u_n]$ and $\mathbf{v} = [v_1, v_2, \ldots, v_n]$, we define the *dot product* $\mathbf{u} \cdot \mathbf{v}$ as

$$\mathbf{u} \cdot \mathbf{v} = (u_1 \times v_1) + (u_2 \times v_2) + \cdots + (u_n \times v_n)$$

The output of the dot product is a *scalar* quantity.

For example,

$$[2, 4, -3] \cdot [5, -1, 1] = 2 \times 5 + 4 \times -1 + -3 \times 1 = 3$$

This may not seem like a very powerful idea, but dot products are *incredibly* useful. The enormous GPUs (Graphics Processing Units) that let video games render scenes so quickly? They primarily function by computing huge numbers of dot products at mind-boggling speeds.

Exercise 1 Basic dot products

Compute the dot product of each pair of vectors:

- $[1, 2, 3]$, $[4, 5, -6]$
- $[\pi, 2\pi]$, $[2, -1]$
- $[0, 0, 0, 0]$, $[10, 10, 10, 10]$

Working Space

Answer on Page 63

2.1 Properties of the dot product

Sometimes we need an easy way to say "The vector of appropriate length is filled with zeros." We use the notation $\vec{0}$ to represent this. Then, for any vector $\mathbf{v}$, this is true:

$$\mathbf{v} \cdot \vec{0} = 0$$

The dot product is commutative:

$$\mathbf{v} \cdot \mathbf{u} = \mathbf{u} \cdot \mathbf{v}$$

The dot product of a vector with itself is its magnitude squared:

$$\mathbf{v} \cdot \mathbf{v} = |\mathbf{v}|^2$$

If you have a scalar a, then:

$$(\mathbf{v}) \cdot (a\mathbf{u}) = a(\mathbf{v} \cdot \mathbf{u})$$

So, if $\mathbf{v}$ and $\mathbf{w}$ are vectors that go in the same direction,

$$\mathbf{v} \cdot \mathbf{w} = |\mathbf{v}||\mathbf{w}|$$

If $\mathbf{v}$ and $\mathbf{w}$ are vectors that go in opposite directions,

$$\mathbf{v} \cdot \mathbf{w} = -|\mathbf{v}||\mathbf{w}|$$

If $\mathbf{v}$ and $\mathbf{w}$ are vectors that are perpendicular to each other, their dot product is zero:

$$\mathbf{v} \cdot \mathbf{w} = 0$$

2.2 Cosines and dot products

Furthermore, dot products' interaction with cosine makes them even more useful is what makes them so useful: If you have two vectors $\mathbf{v}$ and $\mathbf{u}$,

$$\mathbf{v} \cdot \mathbf{u} = |\mathbf{v}||\mathbf{u}| \cos \theta$$

where θ is the angle between them.

So, for example, if two vectors $\mathbf{v}$ and $\mathbf{u}$ are perpendicular, the angle between them is $\pi/2$. The cosine of $\pi/2$ is 0. The dot product of any two perpendicular vectors is always 0. In fact, if the dot product of two non-zero vectors is 0, the vectors *must be* perpendicular (see figure 2.1 for an example of perpendicular 2-dimensional vectors).

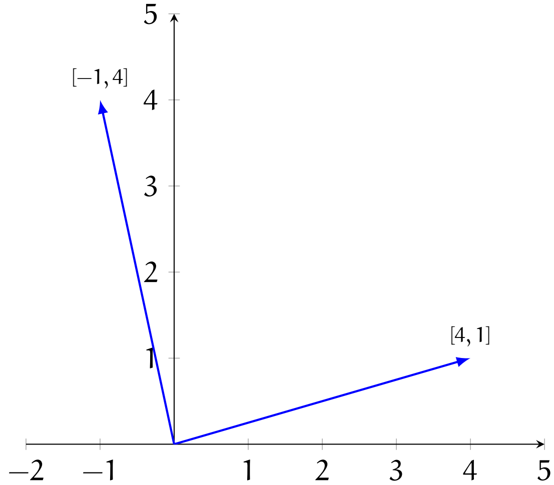


Figure 2.1: The dot product of any two perpendicular vectors is zero.

If you have two non-zero vectors $\mathbf{v}$ and $\mathbf{u}$, you can always compute the angle between them:

$$\theta = \arccos\left(\frac{\mathbf{v} \cdot \mathbf{u}}{|\mathbf{v}||\mathbf{u}|}\right)$$

Arccos is short for arccosine, or $\cos^{-}1$, and it is a function that is the inverse of cosine. Cosine takes an angle and gives back the scaled x-component of the angle. Arccosine takes the x-component of an angle and returns an angle with that x-component. However, there is a limit to what arccos can return. Let's look at cosine and its inverse, arccos (see figures 2.2 and 2.3).

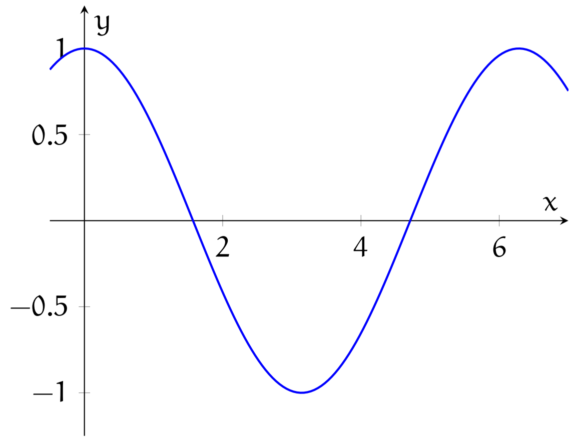


Figure 2.2: Cosine is a function: there is exactly one output for every input.

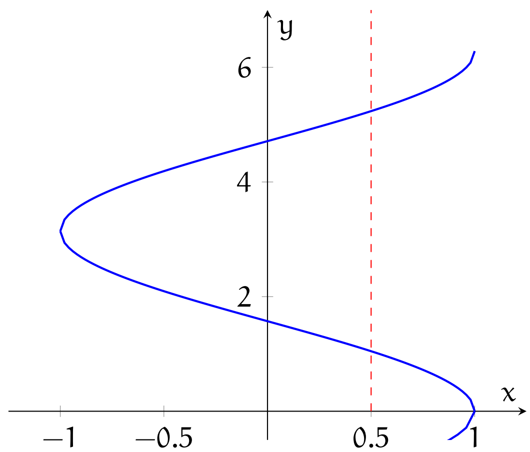


Figure 2.3: Arccos is not a function: there are many angles with the same x-component. Notice that one input value has many output values (see the red dashed line).

When you use a calculator to evaluate arccos, the calculator automatically restricts the results to between 0 and π. Let's look at an example of using the dot product to determine the angle between two vectors:

Example: What is the angle between $\mathbf{u} = \left[\sqrt{3}, 1\right]$ and $\mathbf{v} = [0, -1]$?

Solution: We know that $\mathbf{u} \cdot \mathbf{v} = |\mathbf{u}|\,|\mathbf{v}| \cos\theta$. Therefore, we also know that:

$$\cos\theta = \frac{\mathbf{u} \cdot \mathbf{v}}{|\mathbf{u}|\,|\mathbf{v}|}$$

First, let's compute the dot product:

$$\mathbf{u} \cdot \mathbf{v} = \sqrt{3} \cdot 0 + 1 \cdot -1 = -1$$

And therefore:

$$\cos\theta = \frac{-1}{|\mathbf{u}|\,|\mathbf{v}|}$$

Now, let's find the magnitudes of both vectors:

$$|\mathbf{u}| = \sqrt{\left(\sqrt{3}\right)^2 + (1)^2} = 2$$

$$|\mathbf{v}| = \sqrt{(0)^2 + (-1)^2} = 1$$

Substituting for the magnitudes, we find that:

$$\cos\theta = \frac{-1}{2 \cdot 1} = \frac{-1}{2}$$

To solve for θ, we take the arccos of both sides:

$$\arccos\left(\cos\theta\right) = \theta = \arccos\frac{-1}{2}$$

What angles have a cosine of $-1/2$? We know that $2\pi/3$, $4\pi/3$, $8\pi/3$, etc., all have a cosine of $-1/2$. Because the range of arccos is restricted to between 0 and π, our result is:

$$\theta = \arccos\frac{-1}{2} = \frac{2\pi}{3}$$

.

Therefore, the angle between $\mathbf{u}$ and $\mathbf{v}$ is $2\pi/3$ (or 120°).

Exercise 2 Using dot products

What is the angle between these each pair of vectors:

- $[1, 0]$, $[0, 1]$
- $[3, 4]$, $[4, 3]$
- $[2, -1, 2]$, $[-1, 2, -2]$
- $[-5, 0, -1]$, $[2, 3, -4]$

Working Space

Answer on Page 63

2.3 Dot products in Python

NumPy will let you do dot products using the the symbol @. Open `first_vectors.py` and add the following to the end of the script:

```
# Take the dot product
    d = v @ u
    print("v @ u =", d)

    # Get the angle between the vectors
    a = np.arccos(d / (mv * mu))
    print(f"The angle between u and v is {a * 180 / np.pi:.2f} degrees")
```

When you run it you should get:

```
v @ u = 4
The angle between u and v is 78.55 degrees
```

2.4 Work and Power

Earlier, we mentioned that mechanical work is the product of the force you apply to something and the amount it moves. For example, if you push a train with a force of 10 newtons as it moves 5 meters, you have done 50 joules of work.

What if you try to push the train sideways? It moves down the track 5 meters, but you push it as if you were trying to derail it — perpendicular to its motion. You have done no work, because the train didn't move at all in the direction you were pushing.

Now that you know about dot products: The work you do is the dot product of the force vector you apply and the displacement vector of the train. (The displacement vector is the vector that tells how the train moved while you pushed it.)

Similarly, we mentioned that power is the product of the force you apply and the velocity of the mass you are applying it to. It is actually the dot product of the force vector and the velocity vector.

For example, if you are pushing a sled with a force of 10 newtons and it is moving 2 meters per second, but your push is 20 degrees off, you aren't transferring 20 watts of power to the sled. You are transferring $10 \times 2 \times \cos(20 \text{ degrees}) \approx 18.8$ watts of power.

Chapter 3

Manufacturing

If you try to think of any man-made object, whether it was made from woods, metals, or plastics, chances are it was produced through a manufacturing process.

Over time, these processes have been refined to be more efficient, cost-effective, and faster at producing the goods that we use on a daily basis.

New methods are also constantly being developed by engineers and scientists, and today the range of options available means that choosing the most appropriate manufacturing method involves finding the sweet spot between cost-effectiveness, yield, and time needed.

3.1 Woods and Metals Processes

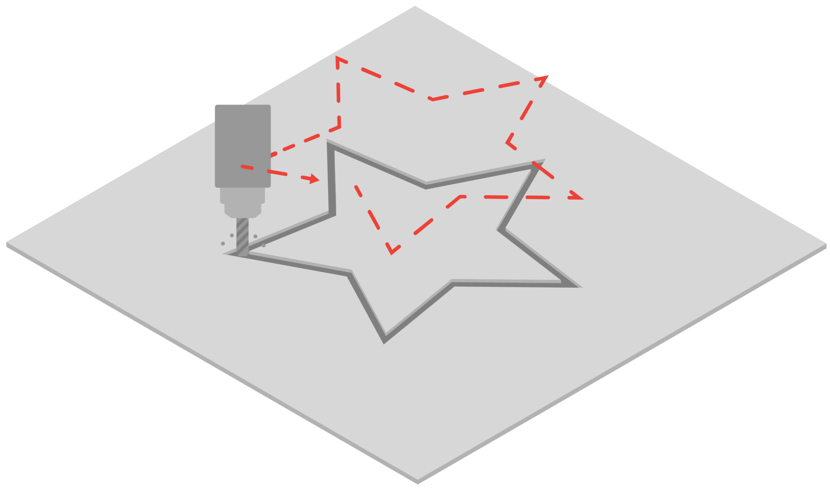
Manufacturing methods that are able to process woods and metals are typically the processes that are used to construct the vast majority of the built world around us.

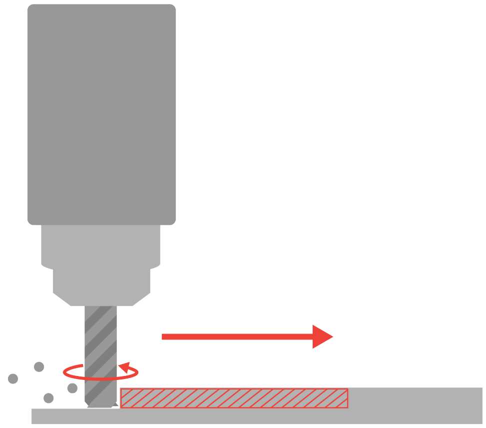
Infrastructure, transportation methods, and buildings would not exist without the advent of processes that allow us to accurately machine raw woods and metals into our desired forms.

The following sections will cover methods used to process both material types.

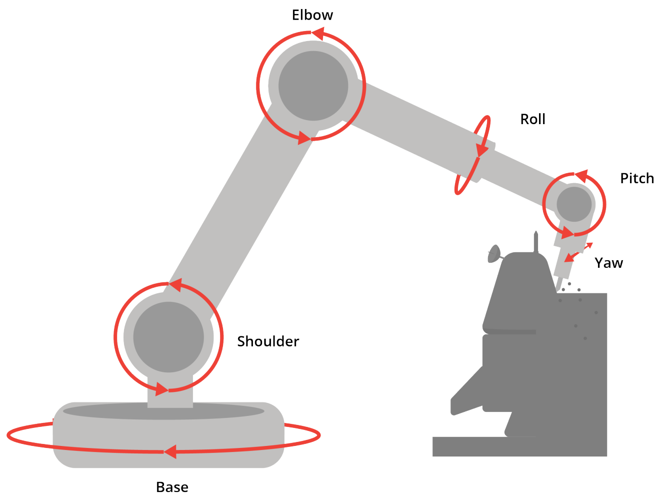
3.1.1 Milling

Mills are powerful tools that allow us to carve out complex shapes from blocks of raw material. A tool bit follows a path to remove the desired material, which makes it a *subtractive* manufacturing process. The tool bit rotates at a very high speed, which allows it to process harder materials such as woods and metals.





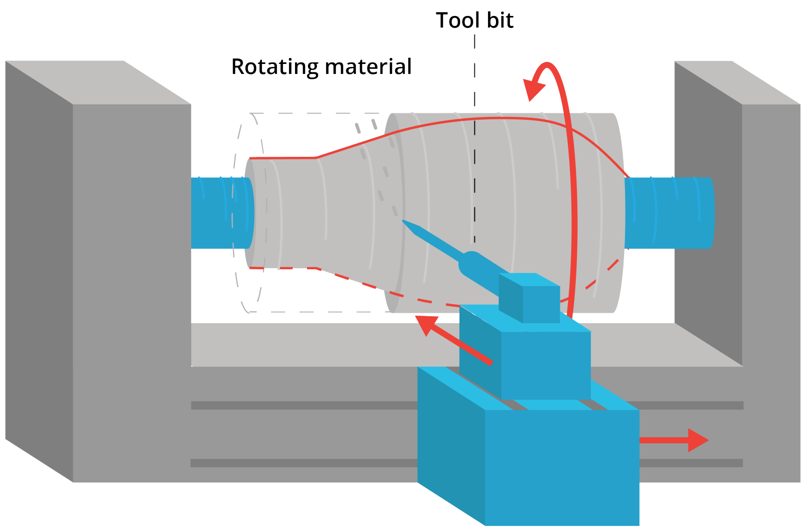
There are various types of mills, ranging from a 3-axis mill, which can cut out simple shapes in the X, Y, and Z axes, all the way to 6-axis mills, which can also rotate about those axes to create more complex curvatures.



In manufacturing settings where speed and repeatability is paramount, mills are often computer controlled. This functionality is referred to as *Computer Numerical Control*, or CNC. CNC mills are able to repeatedly follow a tool path, resulting in consistent and accurate parts.

3.1.2 Lathing

Lathes are tools that allow us to carve out complex cylindrical shapes from raw material. Like a mill, it also is a subtractive manufacturing process. However, lathes rotate the material itself at a high speed, rather than the tool bit. As the material rotates, the tool bit can be used to extract material layer by layer.



In the manufacturing industry, lathes are also often computer controlled. Alongside CNC mills, these CNC lathes are responsible for a majority of the objects that we interact with everyday.

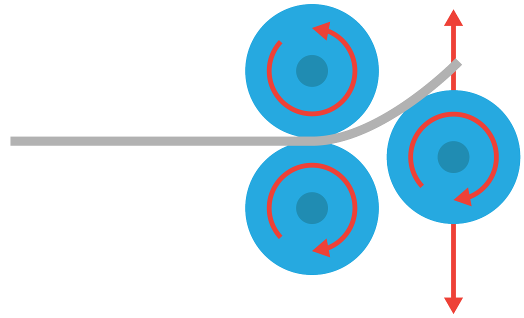
3.2 Metal-specific Processes

While mills and lathes are already able to cover the vast majority of manufacturing needs for woods and metals, there are certain processes that are specifically enabled by the unique properties of metals. More often than not, these processes leverage the malleability (ability to bend without breaking) of metals at room temperature or higher.

3.2.1 Sheet Metal Processes

While milling allows us to process blocks of metal to great effect, sometimes the application we need does not require material of such thickness.

This is where sheet metal comes in, as well as the methods we use to process it. One of the most common techniques in manufacturing is rolling, where a raw sheet is gradually rolled into the desired shape. This method is used to create many objects you may recognize, such as metal roofings, aircraft frames, and more.

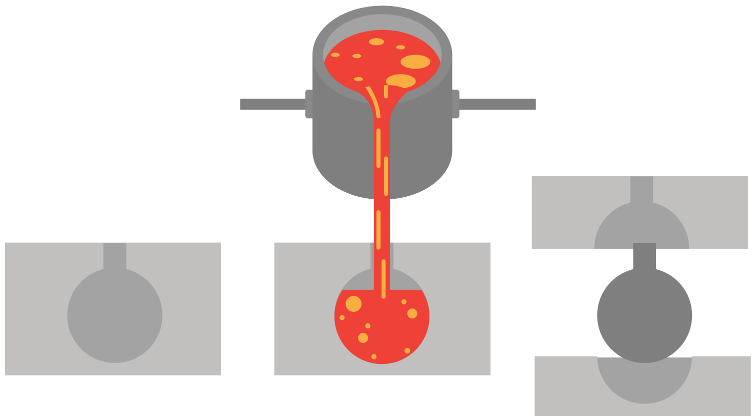


Another method is stamping, where a raw sheet is stamped into the desired shape. This allows us to create surfaces with complex geometries in an instant, and in large quantities. This method is often used for applications like the exterior panels of a car, where parts with compound curves are needed.

Lastly, there are also separate processes used to increase the strength of sheet metal parts. This falls under the category of sheet metal forming, and involves bending the edges of a part to form a reinforced flap. Almost all sheet metal parts are reinforced in this manner, as it is a relatively simple process and also helps to create a clean edge for a more finished look.

3.2.2 Casting

The last kind of metal-specific process we will cover is casting. Casting involves pouring molten metal into a mold, then letting the metal cool and set inside the mold. Smaller components with complex geometries and limited structural requirements (such as toys) are often cast, as it is an accurate and high-volume manufacturing method.



Casting also results in minimal material wastage, as it is not a subtractive manufacturing method where excess material is machined away, but rather only the specific amount of material required is poured in each time.

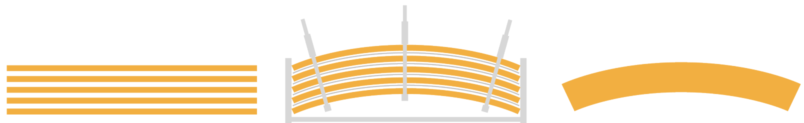
3.3 Wood-specific Processes

Similar to metals, there are also certain manufacturing processes that are enabled by the unique qualities of wood. These make use of the water content inherent in wood, and the flexibility it enables.

3.3.1 Bending

Turning raw wood into flat, workable pieces involves a variety of tools that you probably know of already, such as saws and drills. However, there are specific processes that help us create curved shapes with wood, in addition to the mill and the lathe mentioned earlier.

This is where bent lamination comes in. Bent lamination involves layering multiple thin veneers or strips of wood with adhesive, and clamping it to create the desired form while letting the glue dry. This method is often used for furniture production, enabling continuous curves in wood with tight radii.



Steam can also be used for bending, by helping soften the wood fibers to increase flexibility. Once the desired form is reached, the part can then cool down and harden. Unlike bent lamination, steam bending can be done without adhesives.

3.4 Plastic-specific Processes

Although plastics only came into prominence in the mid 20th century, they have changed manufacturing and, by extension, the world as we know it. Easily manufacturable, durable, and cost-effective, they have come to permeate almost everything we use on a daily basis.

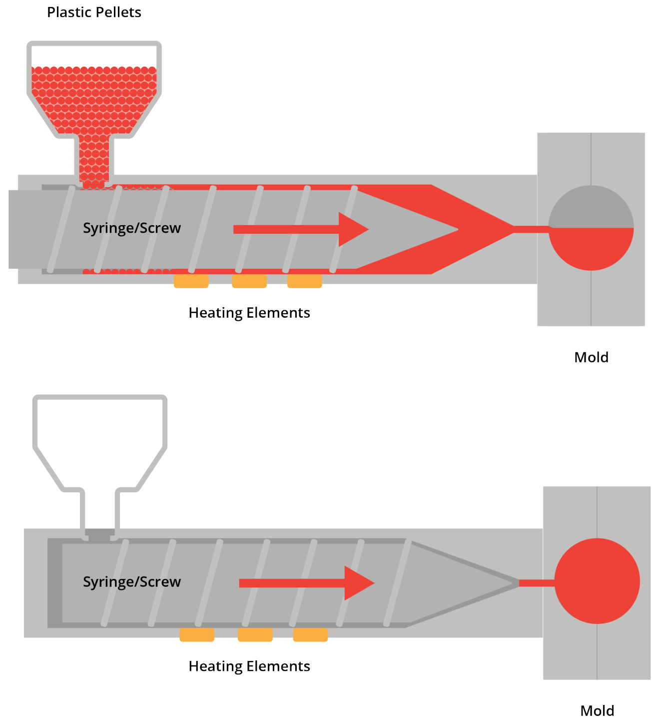
It must also be noted that these exact qualities have also resulted in the proliferation of plastics in our environment, and as such, usage of plastics should be well considered and limited. Alternative biodegradable materials are currently being trialed by scientists and would look to replicate many of the same qualities we expect from plastics, including its manufacturability.

3.4.1 Injection Molding

Injection molding is responsible for the vast majority of plastic products that you interact with on a daily basis. It is extremely quick, highly accurate, and has minimal material wastage, making it a popular and cost effective method of manufacturing plastic goods.

Similar to casting, injection molding involves injecting molten plastic into a mold, then allowing the part to cool and set inside the cavity.

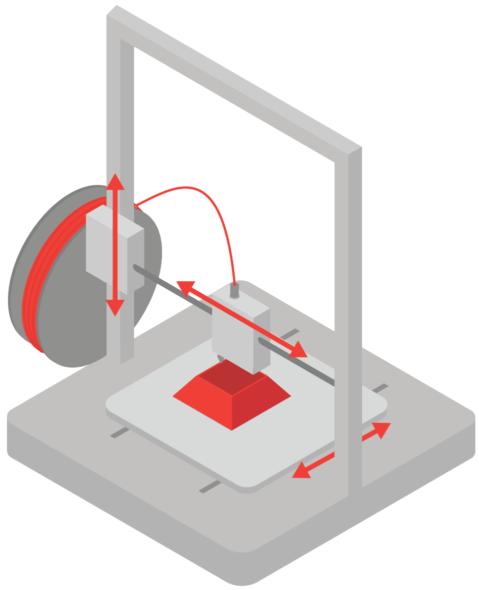
You can often tell when a part was produced through injection molding, with telltale signs such as the parting line. This is where the parts of the mold meet, forming a visible line on the surface of a part.



3.5 3D Printing

Injection molding has traditionally been the go-to technique for manufacturing plastic goods. However, new technologies result in the advent of new manufacturing methods. 3D printing is one such method, having come to prominence in the last few decades as a way to quickly prototype parts without having to create the molds needed for injection molding.

3D printing is an *additive* manufacturing process, where molten material is applied layer by layer to form the desired geometry. It allows for complex geometries, standardized batch production, and whilst the accuracy may currently lag behind traditional injection molding, it is also improving rapidly.

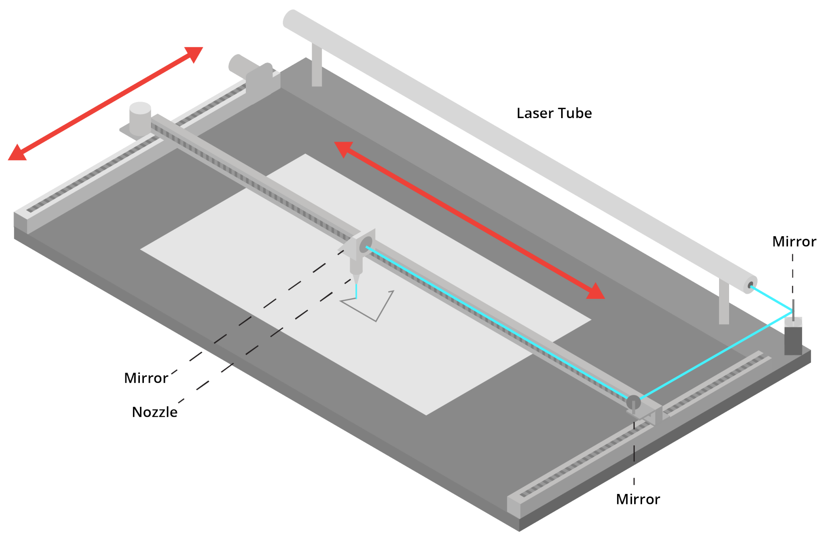


3.6 Laser Cutting and Water Jet

Similarly, another manufacturing method enabled by new technologies is laser cutting. Like 3D printing, it has come into prominence as a method to quickly prototype parts. However, it is not an additive manufacturing process.

Instead laser cutting uses a laser beam to cut and etch through sheets of plastic, however it can also be used for other materials such as fabrics and card stocks. Laser cutting is mostly limited by material thickness, and as such can only cut through thinner sheets of material.

Similarly, water jetting uses pressurized water to blast a stream of water through material (metal, wood, stone, or rubber usually). As the nozzle moves, the high pressure water traces a path throughout the material. Water jets are also limited to material thickness, as too thick of material may not easily be cut.


Laser Tube
Mirror
Mirror
Nozzle
Mirror

Chapter 4

Sound

When you set off a firecracker, it makes a sound.

Let's break that down a little more. Inside the cardboard wrapper of the firecracker, there is potassium nitrate (KNO_3), sulfur (S), and carbon(C). These are all solids. When you trigger the chemical reactions with a little heat, these atoms rearrange themselves to be potassium carbonate (K_2CO_3), potassium sulfate (K_2SO_4), carbon dioxide (CO_2), and nitrogen (N_2). Note that the last two are gasses.

The molecules of a solid are much more tightly packed than the molecules of a gas. So after the chemical reaction, the molecules expand to fill a much bigger volume. The air molecules nearby get pushed away from the firecracker. They compress the molecules beyond them, and those compress the molecules beyond them.

This compression wave radiates out as a sphere; its radius growing at about 343 meters per second ("The speed of sound").

The energy of the explosion is distributed around the surface of this sphere. As the radius increases, the energy is spread more and more thinly around. This is why the firecracker seems louder when you are closer to it. (If you set off a firecracker in a sewer pipe, the sound will travel much, much farther.)

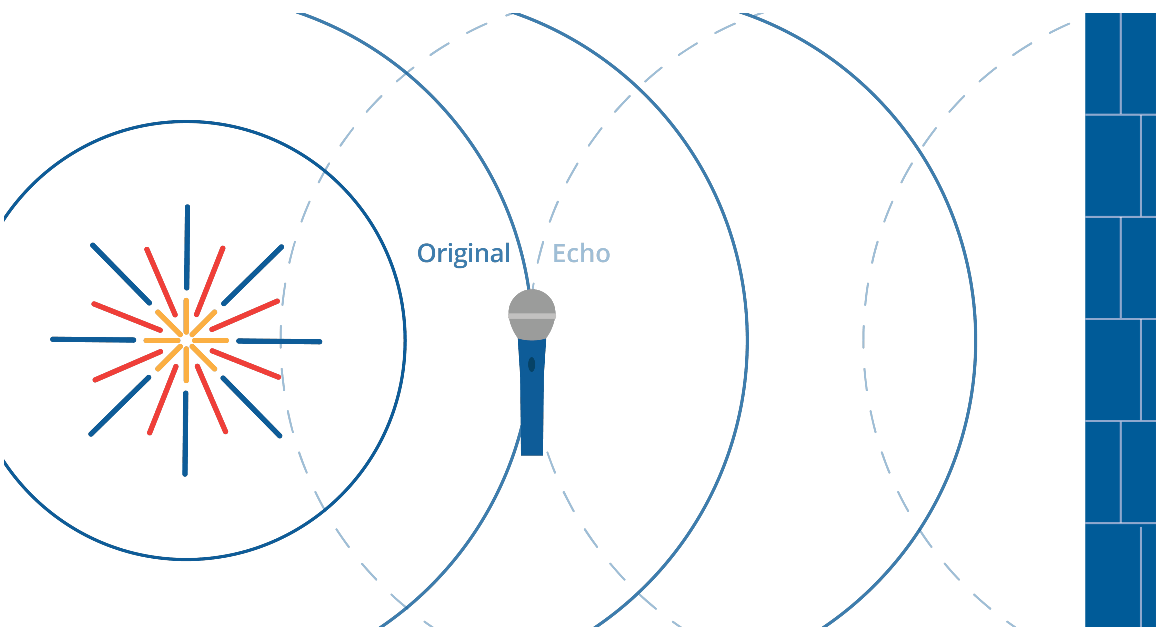


Figure 4.1: A firecracker exploding causes an initial sound wave and an echo.

This compression wave will bounce off of hard surfaces. If you set off a firecracker 50 meters from a big wall, you will hear the explosion twice. We call the second one an "echo".

The compression wave will be absorbed by soft surfaces. If you covered that wall with pillows, there would be almost no echo.

The study of how these compression waves move and bounce is called *acoustics*. Before you build a concert hall, you hire an acoustician to look at your plans and tell you how to make it sound better.

4.1 Pitch and frequency

The string on a guitar is very similar to the weighted spring example. The farther the string is displaced, the more force it feels pushing it back to equilibrium (remember the tension force?). Thus, it moves back and forth in a sine wave. (OK, it isn't a pure sine wave, but we will get to that later.)

The string is connected to the center of the boxy part of the guitar, which is pushed and pulled by the string. That creates compression waves in the air around it.

If you are in the room with the guitar, those compression waves enter your ear and push and pull your eardrum, which is attached to bones that move a fluid that tickles tiny hairs, called *cilia*, in your inner ear. This is how you hear.

We sometimes see plots of sound waveforms. The x-axis represents time. The y-axis represents the amount the air is compressed at the microphone that converted the air pressure into an electrical signal.

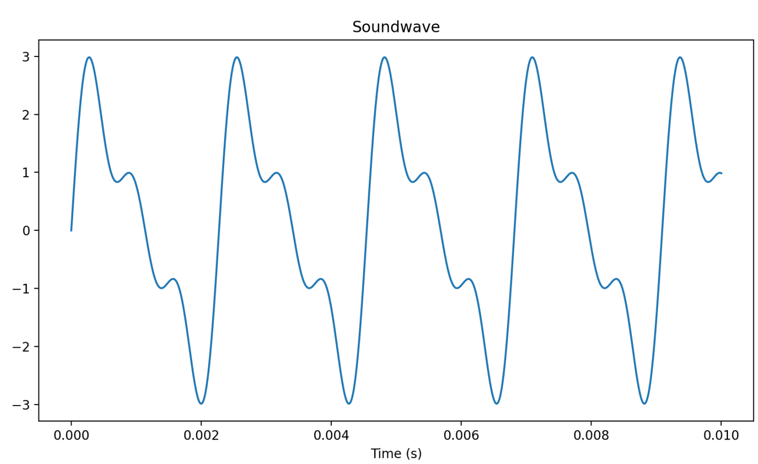


Figure 4.2: A soundwave graph over time.

If the guitar string is made tighter (by the tuning pegs) or shorter (by the guitarist's fingers on the strings), the string vibrates more times per second. We measure the number of waves per second and we call it the *frequency* of the tone. The unit for frequency is *Hertz*: cycles per second. The *period* is opposite of frequency: it is the time it takes for one cycle to complete. The unit for period is seconds per cycle.

Musicians have given the different frequencies names. If the guitarist plucks the lowest note on his guitar, it will vibrate at 82.4 Hertz. The guitarist will say "That pitch is low E." If the string is made half as long (by a finger on the 12th fret), the frequency will be twice as fast (164.8 Hertz), and the guitarist will say "That is E an octave up."

For any note, the note that has twice the frequency is one octave up. The note that has half the frequency is one octave down.

The octave is a very big jump in pitch, so musicians break it up into 12 smaller steps. If the guitarist shortens the E string by one fret, the frequency will be $82.4 \times 1.059463 \approx 87.3$ Hertz.

Shortening the string one fret always increases the frequency by a factor of 1.059463. Why?

Because $1.059463^{1}2 = 2$. That is, if you take 12 of these hops, you end up an octave higher. This, the smallest hop in western music, is referred to as a *half step*.

Exercise 3 Notes and frequencies

Working Space

The note A near the middle of the piano, is 440Hz. The note E is 7 half steps above A. What is its frequency?

Answer on Page 63

4.2 Chords and harmonics

Of course, a guitarist seldom plays only one string at a time. Instead, they use the frets to pick a pitch for each string and strums all six strings.

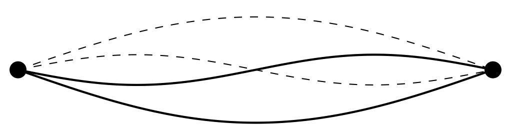
Some combinations of frequencies sound better than others. We have already talked about the octave: If one string vibrates twice for each vibration of another, they sound sweet together.

Musicians speak of “the fifth”. If one string vibrates three times and the other vibrates twice in the same amount of time, they sound sweet together.

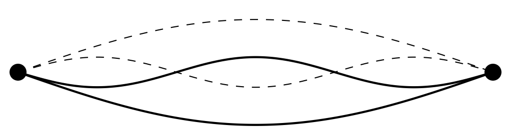
Likewise, if one string vibrates 4 times while the other vibrates 3 times, they sound sweet together. Musicians call this “the third.”

Each of these different frequencies tickle different cilia in the inner ear, so you can hear all six notes at the same time when the guitarist strums their guitar.

When a string vibrates, it doesn’t create a single sine wave. Yes, the string vibrates from end-to-end, and this generates a sine wave at what we call *the fundamental frequency*. However, there are also “standing waves” on the string. One of these standing waves is still at the centerpoint of the string, but everything to the left of the centerpoint is going up, while everything to the right is going down. This creates *an overtone* that is twice the frequency of the fundamental.



The next overtone has two still points — it divides the string into three parts. The outer parts are up, while the inner part is down. Its frequency is three times the fundamental frequency.



And so on. 4 times the fundamental, 5 times the fundamental, etc.

In general, tones with many overtones tend to sound bright. Tones with just the fundamental sound thin.

Humans can generally hear frequencies from 20Hz to 20,000Hz (or 20kHz). Young people tend to be able to hear very high sounds better than older people.

Dogs can generally hear sounds in the 65Hz to 45kHz range.

4.3 Making waves in Python

Let's make a sine wave and add some overtones to it. Create a file named `harmonics.py`.

```
import matplotlib.pyplot as plt
import math

# Constants: frequency and amplitude
fundamental_freq = 440.0 # A = 440 Hz
fundamental_amp = 2.0

# Up an octave
first_freq = fundamental_freq * 2.0 # Hz
first_amp = fundamental_amp * 0.5

# Up a fifth more
second_freq = fundamental_freq * 3.0 # Hz
second_amp = fundamental_amp * 0.4

# How much time to show
max_time = 0.0092 # seconds

# Calculate the values 10,000 times per second
time_step = 0.00001 # seconds

# Initialize
time = 0.0
times = []
totals = []
fundamentals = []
firsts = []
seconds = []

while time <= max_time:
    # Store the time
    times.append(time)

    # Compute value each harmonic
    fundamental = fundamental_amp * math.sin(2.0 * math.pi * fundamental_freq *
    ↪  time)
    first = first_amp * math.sin(2.0 * math.pi * first_freq * time)
    second = second_amp * math.sin(2.0 * math.pi * second_freq * time)

    # Sum them up
    total = fundamental + first + second

    # Store the values
    fundamentals.append(fundamental)
    firsts.append(first)
    seconds.append(second)
    totals.append(total)
```

```

    # Increment time
    time += time_step

# Plot the data
fig, ax = plt.subplots(2, 1)

# Show each component
ax[0].plot(times, fundamentals)
ax[0].plot(times, firsts)
ax[0].plot(times, seconds)
ax[0].legend()

# Show the totals
ax[1].plot(times, totals)
ax[1].set_xlabel("Time (s)")

plt.show()
```

When you run it, you should see a plot of all three sine waves and another plot of their sum:

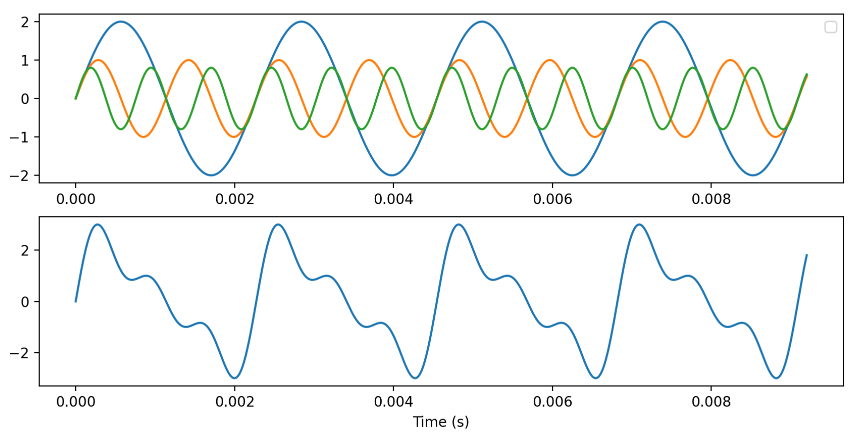


Figure 4.3: The output of haronics.py

4.3.1 Making a sound file

The graph is pretty to look at, but make let's a file that we can listen to.

The WAV audio file format is supported on pretty much any device, and a library for writing WAV files comes with Python. Let's write some sine waves and some noise into a WAV file.

Create a file called `soundmaker.py`

```
import wave
import math
import random

# Constants
frame_rate = 16000 # samples per second
duration_per = 0.3 # seconds per sound
frequencies = [220, 440, 880, 392] # Hz
amplitudes = [20, 125]
baseline = 127 # Values will be between 0 and 255, so 127 is the baseline
samples_per = int(frame_rate * duration_per) # number of samples per sound

# Open a file
wave_writer = wave.open('sound.wav', 'wb')

# Not stereo, just one channel
wave_writer.setnchannels(1)

# 1 byte audio means everything is in the range 0 to 255
wave_writer.setsampwidth(1)

# Set the frame rate
wave_writer.setframerate(frame_rate)

# Loop over the amplitudes and frequencies
for amplitude in amplitudes:
    for frequency in frequencies:
        time = 0.0
        # Write a sine wave
        for sample in range(samples_per):
            s = baseline + int(amplitude * math.sin(2.0 * math.pi * frequency * time))
            wave_writer.writeframes(bytes([s]))
            time += 1.0 / frame_rate

        # Write some noise after each sine wave
        for sample in range(samples_per):
            s = baseline + random.randint(0, 15)
            wave_writer.writeframes(bytes([s]))

# Close the file
wave_writer.close()
```

When you run it, it should create a sound file with several tones of different frequencies and volumes. Each tone should be followed by some noise.

Chapter 5

Applying Numerical Methods to Bivariate Data

5.1 Review of Scatter Plots

5.2 Correlation Coefficient

In the previous section, we used scatterplots to visually judge whether two variables are related. But how do we move beyond ”this looks like a strong relationship” to something we can actually calculate and compare? That is exactly what the *correlation coefficient* gives us.

A correlation coefficient is a numerical measure used to judge the relation between two variables. The one we will use throughout this course is *Pearson's correlation coefficient*, also known simply as the correlation coefficient. It measures the strength and direction of the linear relation between two quantitative variables.

You will encounter two versions of this value depending on whether you are working with a full population or a sample drawn from one. When computed from a population, the correlation coefficient is written as ρ (read as "rho"). When computed from a sample, it is written as r. In this course, we will almost always be working with sample data, so r is the symbol you will use most.

No matter how strong or weak a relationship is, r always falls within a fixed range:

$$-1 \leq r \leq +1$$

Think about what those endpoints mean. A value of $r = +1$ or $r = -1$ means the points in a scatterplot fall perfectly on a straight line. A value of $r = 0$ means there is no linear relationship between the two variables at all. Every other value of r falls somewhere in between, and in the next subsection we will look at exactly how to interpret where r lands.

Exercise 4 Boundaries of r

Answer the following questions about the range of the correlation coefficient.

1. A student computes $r = 1.3$ for a dataset. What does this tell you, without looking at the data?
2. Two researchers study the same population but one uses the full population and the other uses a random sample. What symbol does each researcher use for their correlation coefficient?
3. Can r equal exactly 0.5? Can it equal exactly -1? Explain what each of those values would mean about the data.

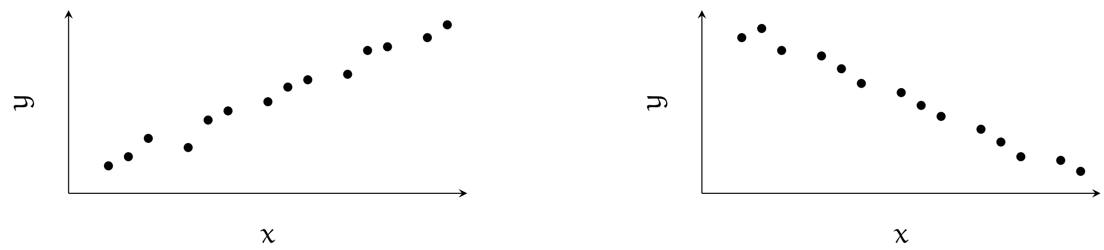
Working Space

Answer on Page 63

Direction

Now that we know r always falls between -1 and $+1$, the first thing to read from it is direction. The sign of r tells you immediately whether the two variables move together or against each other.

A positive value of r indicates that as x increases, y also tends to increase. Taller fathers tend to have taller sons, so the scatterplot shows an upward trend and r is positive.A negative value of r indicates that as x increases, y tends to decrease. As a phone gets older, its battery life tends to decrease, so the scatterplot shows a downward trend and r is negative.



(a) Positive relation ($r > 0$) (b) Negative relation ($r < 0$)

Figure 5.1: Direction of a linear relation as shown in a scatterplot.

Notice that direction says nothing about strength. Both $r = +0.1$ and $r = +0.95$ are positive, but they describe very different relationships. That is what we look at next.

Exercise 5 Reading Direction from r

For each pair of variables, predict whether r would be positive, negative, or zero.

x	y	Sign of r
Hours studied	Exam score	
Temperature	Hot drinks sold	
Shoe size	Intelligence	
Age of a car	Resale value	

Working Space

Answer on Page 64

Strength

Direction tells us which way a relationship goes, but not how strong it is. For that, we look at the numeric value of r, ignoring its sign. The further r is from zero, the stronger the linear relationship between the two variables. The closer r is to zero, the weaker it is.

At the extremes, $r = +1$ or $r = -1$ indicates a *perfect correlation*, meaning every point in the scatterplot falls exactly on a straight line. In practice, real data almost never achieves this. On the other end, $r = 0$ means there is no linear relationship at all.

To compare the strength of two correlations, just compare the absolute values. If $r = -0.86$ for one pair of variables and $r = 0.75$ for another, the first pair has the stronger linear relationship because $0.86 > 0.75$.

Since interpreting strength can be subjective, statisticians often use cutoff values as a rough guide. The diagram below shows one commonly used set of cutoffs. Keep in mind that these are conventions, not rules. The key principle is always the same: the further from zero, the stronger the correlation.

Very Strong	Strong	Weak	No Correlation	Weak	Strong	Very Strong
−1.00 to −0.85	−0.85 to −0.50	−0.50 to −0.10	−0.10 to +0.10	+0.10 to +0.50	+0.50 to +0.85	+0.85 to +1.00

Figure 5.2: One commonly used set of cutoff values for interpreting the strength of a correlation coefficient.

Exercise 6 Interpreting Strength

Working Space

The table below shows correlation coefficients computed from four different datasets. For each one, describe the direction and approximate strength of the relationship.

r	Direction	Strength
+0.92		
−0.34		
+0.07		
−0.88		

Answer on Page 64

Example

Let us put direction and strength together using a real dataset. Consider a sample of 10 father-son pairs. The heights (in inches) are recorded below.

Height of Father x (inches)	Height of Son y (inches)
66	66
66	65
67	67
67	70
68	67
68	70
69	70
70	72
71	72
73	74

Before computing anything, we check whether a correlation coefficient is appropriate. Correlation only applies to linear relationships, so we first make a scatterplot to confirm the relation between the two variables is linear.

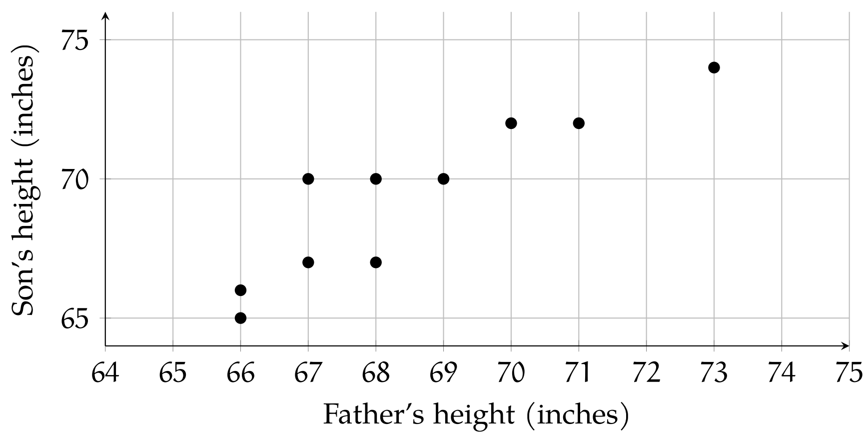


Figure 5.3: Scatterplot of father and son heights for a sample of 12 pairs.

The scatterplot shows a linear trend, so it is appropriate to compute r. In practice, correlation coefficients are computed using a calculator or statistical software rather than by hand. For this dataset, the computed value is approximately $r = 0.96$.

Computing r in Excel

Open your data in Excel or Google Sheets with the x values in column A and the y values in column B. To compute r directly, use the built-in correlation function:

```
=CORREL(A2:A11, B2:B11)
```

This returns the correlation coefficient between the two columns instantly. For the father-son data, this should return approximately 0.96.

Computing r in Python

```python
import numpy as np

# Father and son heights in inches
x = [66, 66, 67, 67, 68, 68, 69, 70, 71, 73]
y = [66, 65, 67, 70, 67, 70, 70, 72, 72, 74]

r = np.corrcoef(x, y)[0, 1]
print(r)
```

`np.corrcoef()` returns a 2×2 matrix of correlation coefficients. The value at position `[0, 1]` is the correlation between x and y. For this dataset, the output should be approximately `0.96`.

Try replacing the data with your own values and observe how r changes as the relationship between the two variables becomes stronger or weaker.

What does this tell us? The sign is positive, so taller fathers tend to have taller sons. The value is 0.96, which falls in the very strong range of our cutoff table. The points cluster tightly around a straight line, and the scatterplot confirms this.

Notice also what r does not tell us. It describes the strength and direction of the linear relationship, but not the shape of any curve, and not whether one variable causes the other. A strong correlation between two variables does not mean one causes the other to change.

Exercise 7 Interpreting a Computed r

Working Space

Answer on Page 64

sectionLinear Regression

Suppose you wanted to predict a teacher's starting salary based on how many years of experience they have. You might expect that more experience leads to a higher salary, but how do you turn that intuition into a precise estimate? That is exactly what a *linear regression model* does. It gives you a straight-line equation that describes the relationship between two variables, so you can use the value of one to predict the value of the other.

The Regression Model

When one variable depends on or is explained by another, we call the variable being predicted the *dependent variable*. The variable doing the explaining is the *independent variable*. In our example, starting salary is the response variable and years of experience is the independent variable, it would make little sense to say that a teacher's salary determines how many years they have worked.

The linear relation between two variables is described by the *population regression equation*:

$$Y = \alpha + \beta X$$

where Y is the response variable, X is the independent variable, α is the y-*intercept* (the value of Y when $X = 0$), and β is the *slope* (the amount Y changes for every one-unit increase in X). This equation describes the true relationship in the population, but in practice, we never know α and β exactly. We estimate them from sample data.

Once we have estimates for α and β, we can write the *estimated regression line*:

$$\hat{y} = a + bx$$

The symbol $\hat{y}$ (read "y-hat") is the *predicted value* of Y for a given value of X. Notice the difference in notation: α and β are population parameters; a and b are their sample-based estimates.

Now, no data will ever fall perfectly on a straight line. A student who studies ten hours might score differently than the line predicts. The difference between what the model predicts and what is actually observed is called the *residual*, or *error*, and is denoted e:

$$e = y - \hat{y}$$

A positive residual means the observed value was higher than predicted; a negative residual means it was lower. Going back to our teachers: if the model predicts a starting salary of \$38,000 for a teacher with four years of experience, but that teacher actually earns \$40,500, the residual is $40{,}500 - 38{,}000 = 2{,}500$. This teacher earned \$2,500 more than the model expected.

Exercise 8 Computing a Residual

A regression model predicts that a teacher with seven years of experience will have a starting salary of \$42,300. The teacher's actual starting salary is \$40,100.

1. Compute the residual.
2. Was the teacher's salary over-predicted or under-predicted? Explain.

Working Space

Answer on Page 64

The goal of linear regression is to find the line that makes these residuals as small as possible across all observations.

The Least-Squares Regression Line

Every straight line you could draw through a scatterplot will produce a different set of residuals. Some lines will over-predict for certain point and under-predict for others. We want to know: which line does the best job overall? The answer is the *least-squares regression line*, also called the *line of best fit*. It is the line that minimizes the sum of the squared residuals, so, it makes $\sum e^2 = \sum (y - \hat{y})^2$ as small as possible.

Why should we square the residuals? First, squaring eliminates the sign, so positive and negative errors do not cancel each other out. Second, squaring penalizes large errors more heavily than small ones, which pushes the line toward a better overall fit. Now you know!

Exercise 9 Why Square the Residuals?

Suppose a regression line produces the following residuals for five data points: $3, -3, 2, -1, -1$.

1. Compute the sum of the residuals. What do you notice?
2. Compute the sum of the squared residuals.
3. Explain why simply minimizing the sum of the residuals would not be a useful goal.

Working Space

Answer on Page 64

The least-squares regression line has two properties that always hold, regardless of the dataset.

First, it always passes through the point $(\bar{X}, \bar{Y})$ which represent the means (or average points) of the two variables. This point is sometimes called the *centroid* of the data.

Second, the slope b of the line is always given by:

$$b = r\left(\frac{S_y}{S_x}\right)$$

where r is the correlation coefficient, S_y is the standard deviation of the Y-values, and S_x is the standard deviation of the X-values. Notice that the slope depends directly on r: if there is no linear relationship ($r = 0$), the slope is zero, and the best prediction for any X is simply $\bar{Y}$.

Once you have b, the y-intercept a follows from the fact that the line passes through $(\bar{X}, \bar{Y})$:

$$a = \bar{Y} - b\bar{X}$$

These two formulas together give you the equation of the least-squares regression line:

$$\hat{y} = a + bx$$

Exercise 10 Computing the Line

For a dataset of teachers' years of experience (X) and starting salary in thousands of dollars (Y), the following summary statistics were computed: $\bar{X} = 6$, $\bar{Y} = 44$, $S_x = 3.2$, $S_y = 5.8$, $r = 0.91$.

1. Compute the slope b of the least-squares regression line.
2. Compute the y-intercept a.
3. Write the equation of the least-squares regression line.
4. The line must pass through the centroid of the data. Verify this using your equation.

Working Space

Answer on Page 65

Interpreting Slope and Intercept

Having the equation of the least-squares regression line is only half the job. The other half is being able to explain what it means in plain language.

The slope b tells you how much the predicted value of Y changes for every one-unit increase in X. The key phrase you could see on an exam is **on average** because b is an estimate based on sample data, not an exact rule.

For our teachers example, if $b = 1.65$, the correct interpretation is:

> For every additional year of experience, the predicted starting salary increases by $1,650, on average.

Notice three things about that sentence. It names both variables. It gives the direction and amount of change. And it includes "on average." Leave out any one of those and the interpretation is incomplete.

Exercise 11 Interpreting the Slope

A least-squares regression line for predicting a teacher's starting salary (in thousands of dollars) from years of experience has slope $b = 1.49$.

1. Write a correct interpretation of the slope in context.
2. A student writes: "For every extra year of experience, salary goes up by \$1,490." What is wrong with this interpretation?

Working Space

Answer on Page 65

The y-intercept a is the predicted value of Y when $X = 0$. Interpreting it requires care. Sometimes $X = 0$ is meaningful where a teacher with zero years of experience is simply a newly hired teacher. In that case the intercept has a natural interpretation: it is the predicted starting salary for someone entering the profession with no prior experience.

Other times $X = 0$ is outside the range of the data entirely, or physically impossible. In those situations, the intercept is mathematically necessary to define the line but has no useful real-world meaning. You should note that and move on.

Exercise 12 Interpreting the Intercept

Using the regression equation $\hat{y} = 34.10 + 1.649x$, where x is years of experience and $\hat{y}$ is predicted starting salary in thousands of dollars:

1. Write a correct interpretation of the y-intercept in context.
2. Is this interpretation meaningful here? Explain.

Working Space

Answer on Page 65

Worked Example

A random sample of 10 teachers was selected from a large urban school district. Each teacher's years of experience at the time of hiring and their starting salary (in thousands of dollars) were recorded. The data is given in the table below.

Experience (years)	Starting Salary ($1000s)
3	36
8	43
1	32
0	30
5	38
10	47
2	33
7	42
4	37
6	41

Before doing any calculations, look at the data. As years of experience increase, starting salaries appear to rise steadily. There are no obvious outliers, and the values are spread across a reasonable range of experience levels. A scatterplot would confirm whether this relationship is linear and this is! There is a strong positive linear relationship between experience and starting salary.

From the data, we have $n = 10$ pairs of measurements. Using a calculator or software, we can obtain the following summary statistics:

$$\bar{X} = 4.6, \quad \bar{Y} = 37.9, \quad S_x = 3.169, \quad S_y = 5.013, \quad r = 0.9964$$

Exercise 13 Slope, Intercept, and the Regression Equation

Using the summary statistics above:

1. Compute the slope b of the least-squares regression line. Interpret it in context.
2. Compute the y-intercept a. Interpret it in context.
3. Write the equation of the least-squares regression line.

Working Space

Answer on Page 65

Exercise 14 Prediction, Residuals, and R^2

Using the regression equation $\hat{y} = 30.650 + 1.576x$:

1. Predict the starting salary for a teacher with 5 years of experience.
2. The teacher in the sample with 8 years of experience has an actual starting salary of \$43,000. Compute the residual and interpret it.
3. Compute the coefficient of determination R^2 and interpret it in context.
4. About what percentage of the variation in starting salaries is *not* explained by years of experience? Suggest one factor that might account for this.

Working Space

Answer on Page 66

textbfComputing in Excel

You do not need to compute the slope, intercept, and R^2 by hand every time. Excel can produce all of these values at once, along with a scatterplot showing the line of best fit. Here is how.

First, enter your data into two columns, with experience values in column A and starting salary values in column B, with headers in row 1. Select both columns, then insert a scatterplot using **Insert** → **Chart** → **Scatter**. Once the chart appears, click on any data point, then select **Add Trendline**. In the trendline options panel on the right, choose **Linear** and check both **Display Equation on chart** and **Display R-squared value on chart**. Excel will draw the line of best fit and print the equation and R^2 directly on the chart.

To get the full regression output, use the **Data Analysis** Toolpak. Go to **Data** → **Data Analysis** → **Regression**. Set the Input Y Range to your salary column and the Input X Range to your experience column. Check **Labels** if your columns have headers. Click OK and Excel will produce a full regression table in a new sheet. The values you need are in the **Coefficients** column: the row labelled **Intercept** gives you a, and the row labelled

Experience gives you b. The R^2 value appears near the top of the output under **R Square**.

Computing in Python

Python can produce the same regression results as Excel, and a little more besides. The code below uses two libraries: `numpy` for the summary statistics and `scipy` for the regression, and `matplotlib` to draw the scatterplot with the line of best fit.

```
import numpy as np
from scipy import stats
import matplotlib.pyplot as plt

# Data
experience = [3, 8, 1, 0, 5, 10, 2, 7, 4, 6]
salary     = [36, 43, 32, 30, 38, 47, 33, 42, 37, 41]

# Least-squares regression
slope, intercept, r, p, se = stats.linregress(experience, salary)

print(f"Slope (b):     {slope:.3f}")
print(f"Intercept (a): {intercept:.3f}")
print(f"r:             {r:.4f}")
print(f"R-squared:     {r**2:.4f}")

# Scatterplot with regression line
x_line = np.linspace(0, 10, 100)
y_line = intercept + slope * x_line

plt.scatter(experience, salary, color="steelblue", label="Data")
plt.plot(x_line, y_line, color="firebrick",
         label=f"y = {intercept:.2f} + {slope:.2f}x")
plt.xlabel("Experience (years)")
plt.ylabel("Starting Salary ($1000s)")
plt.title("Experience vs. Starting Salary")
plt.legend()
plt.show()
```

Running this code produces the following output:

```
Slope (b):     1.673
Intercept (a): 30.203
r:             0.9957
R-squared:     0.9915
```

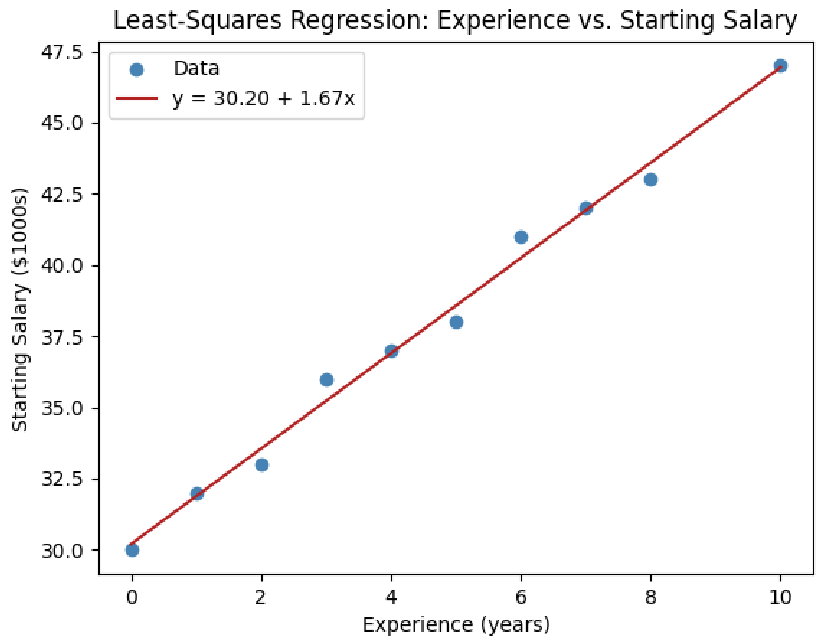


Figure 5.4: Experience vs. Starting Salary Graph generated from Python

These match the values we computed by hand exactly. The `stats.linregress` function does all the heavy lifting! It returns the slope, intercept, correlation coefficient `r`, a p-value, and the standard error of the slope in one line. For now, focus on the first four outputs. The p-value and standard error will become important when we study inference for regression later!

Appendix A

Answers to Exercises

Answer to Exercise 1 (on page 21)

- $[1, 2, 3] \cdot [4, 5, -6] = 4 + 10 - 18 = -4$
- $[\pi, 2\pi] \cdot [2, -1] = 2\pi - 2\pi = 0$
- $[0, 0, 0, 0] \cdot [10, 10, 10, 10] = 0 + 0 + 0 + 0 = 0$

Answer to Exercise 2 (on page 26)

- $[1, 0] \cdot [0, 1] = 0$. The angle must be $\pi/2$.
- $[3, 4] \cdot [4, 3] = 24$. $|[3, 4]||[4, 3]| \cos(\theta) = 24$. $\cos(\theta) = \frac{24}{(5)(5)}$. $\theta = \arccos(\frac{24}{25}) \approx 0.284$ radians.
- $[2, -1, 2] \cdot [-1, 2, -2] = 4 - 2 - 4 = -2$. $|[2, -1, 2]| = \sqrt{4 + 1 + 4} = \sqrt{9} = 3$. $|[-1, 2, -2]| = \sqrt{1 + 4 + 4} = \sqrt{9} = 3$. $3(3) \cos \theta = -2$. $\theta = \arccos(-2/9) \approx 1.795$ radians.
- $[-5, 0, -1] \cdot [2, 3, -4] = -10 + 0 + 4 = -6$. $|[-5, 0, 1]| = \sqrt{25 + 0 + 1} = \sqrt{26}$. $|[2, 3, -4]| = \sqrt{4 + 9 + 16} = \sqrt{29}$. $\sqrt{26}(\sqrt{29}) \cos \theta = -6$. $\theta = \arccos(\frac{-6}{\sqrt{26}\sqrt{29}}) \approx 1.791$ radians.

Answer to Exercise 3 (on page 41)

A is 440 Hz. Each half-step is a multiplication by $\sqrt[12]{2} = 1.059463094359295$ So the frequency of E is $(440)(2^{7/12}) = 659.255113825739859$

Answer to Exercise 4 (on page 48)

1. A value of $r = 1.3$ is impossible. The correlation coefficient must satisfy $-1 \leq r \leq +1$, so the student made a calculation error.
2. The researcher using the full population uses ρ. The researcher using a sample uses

r.

3. Both are possible. An r of 0.5 would indicate a moderate positive linear relationship. The points follow an upward trend but with noticeable scatter. An r of -1 would indicate a perfect negative linear relationship, meaning all points fall exactly on a downward-sloping line.

Answer to Exercise 5 (on page 49)

x	y	Sign of r
Hours studied	Exam score	Positive
Temperature	Hot drinks sold	Negative
Shoe size	Intelligence	Zero
Age of a car	Resale value	Negative

Answer to Exercise 6 (on page 50)

r	Direction	Strength
$+0.92$	Positive	Very strong
-0.34	Negative	Weak
$+0.07$	Positive	No correlation
-0.88	Negative	Very strong

Answer to Exercise 7 (on page 52)

Answer to Exercise 8 (on page 54)

1. $e = 40{,}100 - 42{,}300 = -2{,}200$.

2. The salary was over-predicted. The observed salary was \$2,200 less than the model expected.

Answer to Exercise 9 (on page 55)

1. $3 + (-3) + 2 + (-1) + (-1) = 0$. The positive and negative residuals cancel out,

making the sum zero even though the line does not fit perfectly.

2. $3^2 + (-3)^2 + 2^2 + (-1)^2 + (-1)^2 = 9 + 9 + 4 + 1 + 1 = 24$.
3. A line that over-predicts some values and under-predicts others by equal amounts would have a residual sum of zero but it would be a poor fit. Squaring forces every error to contribute positively, so the minimization is meaningful!

Answer to Exercise 10 (on page 56)

1. $b = 0.91\left(\frac{5.8}{3.2}\right) = 0.91 \times 1.8125 \approx 1.649$
2. $a = 44 - 1.649(6) = 44 - 9.897 \approx 34.103$
3. $\hat{y} = 34.103 + 1.649x$
4. Substituting $x = \bar{X} = 6$: $\hat{y} = 34.103 + 1.649(6) = 34.103 + 9.897 = 44.000 = \bar{Y}$. ✓

Answer to Exercise 11 (on page 57)

1. For every additional year of teaching experience, the predicted starting salary increases by \$1,490, on average.
2. The interpretation is missing the phrase "on average" and the word "predicted." The slope is an estimate from sample data, not a guarantee that every teacher's salary increases by exactly \$1,490 per year.

Answer to Exercise 12 (on page 58)

1. The predicted starting salary for a teacher with zero years of experience is \$34,100.
2. Yes, it is meaningful in this context. A teacher with no prior experience is a realistic case, so $X = 0$ is within the plausible range of the data.

Answer to Exercise 13 (on page 59)

1.
$$b = r\left(\frac{S_y}{S_x}\right) = 0.9964\left(\frac{5.013}{3.169}\right) \approx 1.576$$
For every additional year of teaching experience, the predicted starting salary increases by $1,576, on average.

2.
$$a = \bar{Y} - b\bar{X} = 37.9 - 1.576(4.6) \approx 30.650$$
The predicted starting salary for a teacher with no prior experience is $30,650. Since $X = 0$ represents a newly hired teacher with no experience, this interpretation is meaningful in context.

3. The equation of the least-squares regression line is:
$$\hat{y} = 30.650 + 1.576x$$
where x is years of experience and $\hat{y}$ is the predicted starting salary in thousands of dollars.

Answer to Exercise 14 (on page 60)

1.
$$\hat{y} = 30.650 + 1.576(5) = 30.650 + 7.880 = 38.530$$
The predicted starting salary for a teacher with 5 years of experience is $38,530.

2.
$$\hat{y} = 30.650 + 1.576(8) = 30.650 + 12.608 = 43.258$$
$$e = y - \hat{y} = 43 - 43.258 = -0.258$$
The residual is −$258. The teacher's actual starting salary was $258 less than the model predicted. The salary was slightly over-predicted.

3.
$$R^2 = r^2 = (0.9964)^2 \approx 0.9928$$
About 99.28% of the variation in starting salaries is explained by the linear relationship with years of experience. This is an exceptionally strong fit.

4. About $100 - 99.28 = 0.72\%$ of the variation is unexplained. This is very small, but it may be due to factors such as the teacher's level of education, the subject area taught, or the specific school within the district.

Index